

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #2

Instructions

- TA in charge: Abidur Rahman (abidur.rahman@mail.mcgill.ca)
- Write/type your answers in the space provided.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	
First student ID number:	
Second student full name:	
Second student ID number:	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (12%)	Question 2 (05%)	Question 3 (13%)	Question 4 (10%)	Question 5 (10%)	Total
Question 6 (10%)	Question 7 (10%)	Question 8 (10%)	Question 9 (20%)		

Question 1 (12%)

The K-map of a 4-variable logic function $f(x_1, x_2, x_3, x_4)$ is

		x_1/x_2			
		00	01	11	10
x_3/x_4	00	0	0	1	0
	01	1	1	1	0
	11	0	1	1	1
	10	0	1	0	0

(a) List the prime implicants of the function f

(b) Group the prime implicants from (a) into essential and non-essential prime implicants

(c) Provide $\bar{f}$ (the complement of the function f) in SOP form

Question 2 (05%)

Provide the K-map of the following 5-variable logic function
 $g(x_1, x_2, x_3, x_4, x_5) = x_1'x_3'x_5 + x_1'x_3x_4 + x_1x_2'x_4x_5 + x_2x_4'x_5$

Question 3 (13%)

Consider the following Boolean function:

$$h(x_1, x_2, x_3, x_4) = \sum m(3, 7, 8, 9, 10, 11, 13, 15)$$

Find the simplest SOP of h , draw the circuit using only NAND and AND gates. Assume that the input variables are available in both true and complemented form at no additional cost.

Question 4 (10%)

Consider the function $f = \sum m(2,4,6,9,12) + D(1,3,13,14)$

(a) Find the simplest SOP expression of the function using a K-map (show your work).

(b) Find the simplest POS expression of the function using a K-map (show your work).

Question 5 (10%)

Consider the function $f = \sum m(1,3,4,9,11,13,15) + D(5,6,7,14)$

Find the simplest SOP and POS expressions of the function using K-maps (show your work).

Question 6 (10%)

Given the K-map of a 5-variable function f below:

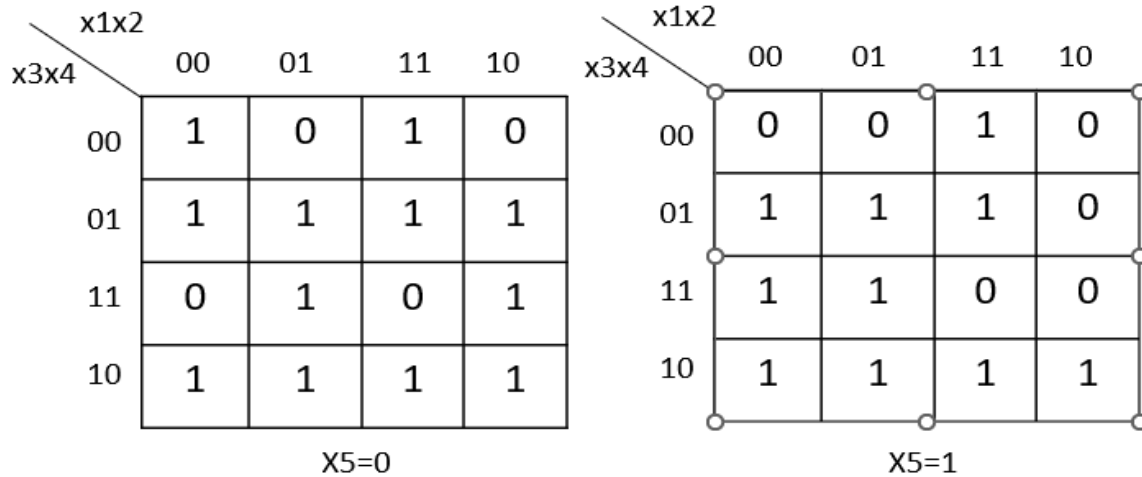
		x1x2						x1x2			
		00	01	11	10			00	01	11	10
x3x4	00	1	1	1	0	x3x4	00	1	1	1	0
	01	0	0	0	1		01	0	0	0	1
	11	1	1	1	0		11	1	1	1	0
	10	0	0	0	1		10	0	0	0	1
x5=0						x5=1					

(a) Find the optimal (lowest cost) multi-level expression of f using functional decomposition.

(b) Draw the circuit implementation from Q.1(a) with maximum fan-in equals to 2.

Question 7 (10%)

Given the K-map of a 5-variable function $f(x_1, x_2, x_3, x_4, x_5)$ below:



Using functional decomposition, find the expression of a multi-level implementation of f .
Hint: There could be more than two patterns in the K-map. Show your work and mark the patterns on the K-map.

Question 8 (10%)

The VHDL code below describes the following system.

Inputs: 4-bit A and 2-bit D. Output: 1-bit Out

Functionality: When D = 00, Out = 0. When D = 01, Out = 1, if A is divisible by 2.

When D = 11, Out=1, if A is divisible by 3, When D = 10, Out=1, if A is divisible by 5. For all other cases, Out = 0.

Fill in the blanks of the provided code accordingly.

```
library IEEE;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity vhdllisawesome is
  port (
    A    : in std_logic_vector(____ downto ____);
    D    : in std_logic_vector(____ downto ____);
    Out : out std_logic);
end vhdllisawesome;

architecture description of _____ is
  signal div_by_2 : std_logic;
  signal div_by_3 : std_logic;
  signal div_by_5 : std_logic;
begin
  with A select div_by_2 <=
    '1' when "0000",
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '0' when OTHERS;

  with A select div_by_3 <=
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '0' when OTHERS;

  with A select div_by_5 <=
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '1' when _____,
    '0' when OTHERS;

  with D select Out <=
    _____ when _____,
    _____ when _____,
    _____ when _____,
    _____ when _____;

end description;
```

Question 9 (20%)

Use separate and joint optimization strategies to find out which one has the lowest-cost SOP implementation of f and g , draw the circuit and provide the cost in each case.

$$f(x_1, x_2, x_3, x_4) = \sum m(0,1,3,5,8,10,11) + D(7,14)$$

$$g(x_1, x_2, x_3, x_4) = \sum m(1,3,5,7,10,11,15) + D(0,9)$$

Note: The complements of the input variables (x_1, x_2, x_3, x_4) are given at no additional cost.

